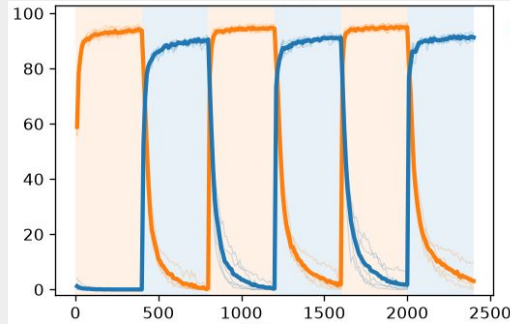


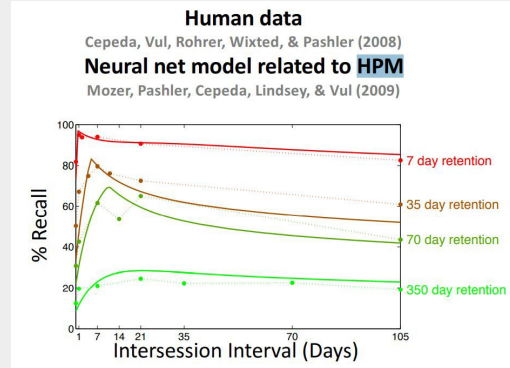
Does energy-based learning
reduce catastrophic forgetting

Artificial Neural Networks forget, humans don't

Machines



Humans



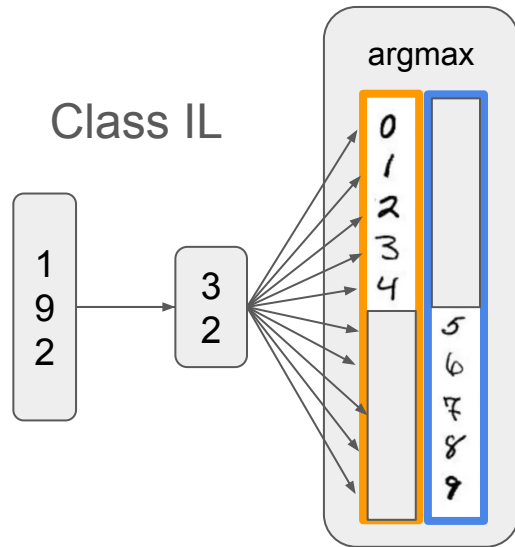
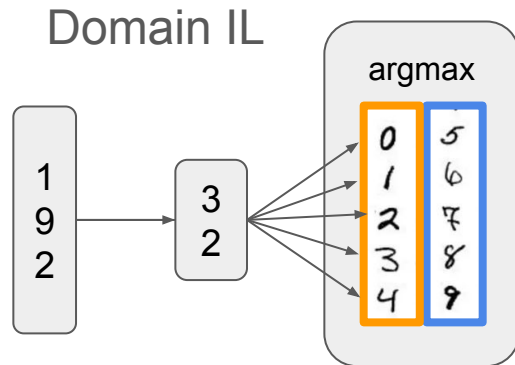
Appropriate capture from neuroscience literature on humans not forgetting?

Then list the forgetting mitigations:
Local learning <- highlight this as focus for EBMs

- Local learning
- Meta-plasticity
- Hippocampal Replay
- And what's the other one? Or are there more?
- This all needs citation.

Architecture and IL Scenarios

- MNIST digits 14x14 (192 units)
- 1 hidden layer (32 units)
- tanh activations, argmax output (classification)
- Incremental learning [IL] **task1** then **task2**
- Randomised task partitions, 5-10 runs



Local learning rule - Prospective configuration [PC]

Does PC forget less?

% Improvement vs Back Propagation

Scenario	Crossover accuracy	Final accuracy	Mean error
Domain IL (fixed)	-0.01	-2.56	+2.97
Domain IL (matched)	+0.49	+2.48	-1.60
Class IL (fixed)	+1.29	-2.04	+3.24
Class IL (matched)	+3.00	+1.60	-5.48